

Requirement Analysis

Solution Requirements (Functional & Non-functional)

Date	29-06-2026
Project Name	Automated Network Request Management
Team ID	
Maximum Marks	

Functional Requirements

These define the core system capabilities and behaviors required for the automated network access management workflow in ServiceNow.

FR no.	Requirement Description	Success / Acceptance Criteria
FR-1	Input Validation & Sanitization	The ServiceNow Service Catalog form must validate network inputs (e.g., IP addresses, ports, protocols) using predefined regex and mandatory field checks before submission.
FR-2	Risk-Based Approval Workflow	Flow Designer must automatically classify requests by risk level and route them accordingly: low-risk requests are auto-approved, while high-risk requests require manager or security approval.

FR-3	Automated API Integration	Integration Hub must convert approved request data into structured JSON/XML payloads and successfully execute secure REST API calls to target network controllers.
FR-4	Deployment Verification	After provisioning, the system must automatically perform connectivity or port verification tests to confirm the requested network changes were applied successfully.
FR-5	CMDB Synchronization & Audit Logging	Successful changes must automatically update relevant Configuration Items (CIs) and relationships in the CMDB while generating immutable audit logs for compliance tracking.
FR-6	Real-Time Notifications	The system must notify stakeholders (requestor, approvers, administrators) at key lifecycle stages such as submission, approval, execution, and closure through email or integrated messaging platforms.

Non-functional Requirements

These define the system's operational attributes—how well the system performs its functions. Since you are building on the ServiceNow platform, you inherit many of these automatically, but they must still be documented.

ID	Requirement Description	Success / Acceptance Criteria

NFR-1	Performance / Speed	Automated request fulfillment (from the moment of final approval to successful API execution) must complete in less than 15 minutes for standard paths.
NFR-2	Security & Authentication	All outbound API calls from ServiceNow Integration Hub to network infrastructure must use encrypted protocols (HTTPS/SSH) and credential aliases stored securely in the platform.
NFR-3	Scalability	The workflow orchestration system must be capable of processing up to 500 concurrent automated network request transactions without causing API timeouts or platform lag.
NFR-4	Availability & Reliability	The automated request engine must achieve 99.9% uptime during operational hours, with graceful error-handling that routes failed API attempts to a manual NetOps queue.
NFR-5	Usability (UX)	Intake forms must use dynamic UI policies to show/hide fields, ensuring a minimalist layout that minimizes user input confusion and reduces incomplete submissions.